The Induction Principle — problem 8.10

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(a) Any simple not necessarily convex n-gon has at least one diagonal which lies completely inside the n-gon.

(b) This n-gon can be triangulated by diagonals which lie inside the n-gon.

(c) The vertices of the triangulated n-gon can be colored properly with three colors.

(d) The faces of the triangulation can be properly colored with two colors.